PRAgent: Claim-level Scientific Reproduction Across 100 Physics Papers

PRAgent Collaboration

27 August 2026

Abstract

Can an AI agent recover the scientific content of a physics paper rather than imitate its figures? We introduce PRAgent, a claim-centred reproduction system, and audit a frozen cohort of 100 physics papers. The evaluation maps 3,933 independently checkable numerical objects to 1,427 scientific Claims. Every Check has one direct Claim and an auditable path to executable targets and evidence. At Claim level, 100.0% of the numerical denominator received a final disposition: 40.5% was successfully reproduced, 21.9% was objectively blocked, 37.5% was attempted but not reproduced, and 0.0% remained pending. Within the reproduced Check mass that has lifecycle-valid scientific-region comparisons, conditional Fidelity was 93.0/100; such evidence covered 23.4% of successful Claim mass. The corresponding conditional plug-in Reproduction Degree was 37.7/100, with non-imputation bounds 8.8–39.9/100. We report the coverage and Fidelity components alongside this optional product so that external blockers are not converted into low pixel scores and sparse high- fidelity successes cannot inflate whole-paper performance. The remaining 15 Claims are evaluated separately by derivation or invariant checks, or excluded when they are experimental or meta-level. These results demonstrate scalable, auditable scientific reproduction while exposing the present limits imposed by missing publication inputs, paper-scale compute, unclosed scientific evidence and unsuccessful implementations.

Significance

Most agent benchmarks test questions, isolated coding tasks or reproduction from supplied repositories. Scientific reproduction is stricter: the agent must identify what a paper claims, reconstruct the governing equations, implement the calculation independently, test physical invariants and only then compare the generated scientific object with the publication. Across 100 papers, PRAgent provides a fixed denominator of 3,933 numerical Checks and a one-to-one direct mapping from every Check to a scientific Claim. The audit does not collapse all failure into one number. It separately reports successful reproduction, objective external obstruction, attempted failure and unfinished work; it assigns no Fidelity to a result without trusted scientific-region comparison evidence. This makes both capability and limitation quantitative, inspectable and suitable for independent challenge.

1 Introduction

Reproducibility has several meanings, ranging from recomputing a result with the original materials to independently reconstructing the method and testing whether the scientific conclusion survives[1]. The latter is an unusually demanding test for an AI agent. Information needed for one result may be distributed across prose, equations, captions and supplementary material. The implementation must choose numerical methods and convergence criteria, while the evaluation must distinguish a code defect from missing publication information or a possible inconsistency in the paper.

Recent systems have shown that language-model agents can plan experiments, operate scientific tools and automate parts of research workflows[2–4]. Reproduction benchmarks have likewise begun to evaluate full research artifacts rather than short answers[5–11]. Physics-specific evaluation adds a further difficulty: agreement must be tied to the intended observable, symmetry, regime and parameter set. Frontier- physics question benchmarks such as PRL-Bench probe physical reasoning but do not by themselves establish end-to-end numerical reproduction [12].

Here we ask a narrower and falsifiable question: *to what degree did an agent independently reproduce the scientific Claims of each paper?* We make three contributions. First, we define a simple claim-first measurement model that separates the scientific denominator from implementation detail. Second, we enforce an evidence boundary in which numerical runners cannot read source figures and pixel comparison cannot modify numerical arrays. Third, we publish the complete Check, Claim and paper ledgers for a fixed 100-paper cohort, including negative outcomes and reasons for every objective blocker.

The central result is not a declaration that all papers are "complete." It is a decomposed capability measurement. At Claim level, successful coverage is 40.5%; conditional Fidelity on the trusted measured subset is 93.0/100. Their optional product is 37.7/100, but uncertainty caused by missing valid Fidelity evidence remains visible as the interval 8.8–39.9/100. This decomposition prevents both optimistic cherry-picking and the misleading assignment of zero visual similarity to work that could not be run for external reasons.

2 System and evidence model

2.1 Claim is the scientific unit; Check is the observable unit

A **Claim** is a scientific statement whose truth can be challenged: for example, a scaling law, a phase boundary, a spectrum, a conserved quantity or an inequality. A **Claim Check** is the smallest independently assessable observable used to test that Claim. It can be a numerical subpanel, a curve, a table entry, a fitted exponent or a formula-level invariant. A Claim may require several Checks; a Check has exactly one direct Claim. Explicit supporting edges may connect the same Check to a higher-level Claim, but they never replace the direct edge and are not inferred from textual or visual similarity.

A **Target** is an execution contract. It defines what code must produce, with which equations, parameters, outputs and scientific tests. Targets are implementation objects and may support several Checks. Keeping Claims and Targets distinct avoids two common errors: treating a plot file as the scientific assertion, and counting one implementation detail as several scientific successes.

2.2 Independent numerical generation

Each numerical Target begins with a formula and parameter contract. The runner executes in an isolated directory and records the Git revision, configuration hash, effective arguments, file-access manifest and output hashes. It may read the paper and parameter evidence made available by the contract, but it cannot read raw source figures or image-derived arrays. Published author code is not used to reconstruct the numerical method; declared parameters and final results may be consulted as publication evidence. The generated data are frozen before any visual optimisation.

Physics checks precede image comparison. Depending on the problem, these may test normalization, conservation laws, limiting cases, Hermiticity, eigenvalue pairing, symmetry sectors, convergence or independently derived analytic relations. A process that merely executes successfully does not therefore count as a scientific reproduction.

2.3 Constrained rendering and fresh-context review

After the numerical arrays are frozen, a RenderContract may use the source figure to adjust only canvas geometry, axes, fonts, line styles, palette and interpolation. It cannot modify physical parameters, reorder scientific data or replace generated arrays with sampled source pixels. Pixel comparison is restricted to predeclared primary scientific regions; full-frame similarity is retained only as a layout diagnostic.

Finally, a fresh-context reviewer receives the paper, frozen formula/parameter scope, implementation and generated evidence, but not the original reproduction conversation or its conclusions. The reviewer independently enumerates the numerical scope, attempts to falsify the results and classifies discrepancies as supported, inconclusive, reproduction defect or paper-error candidate. Authoritative state is then derived mechanically from the evidence; the reproducing agent cannot promote its own result to completion.

Table 1: **Frozen claim-first audit.** Rates at Claim level give every numerical Claim equal weight; Check-level rates expose the underlying atomic workload. Fidelity is conditional on successful items with valid evidence.

Metric (%)	Check level	Claim level
Final resolution	100.0	100.0
Successful reproduction	52.6	40.5
Objective blocker	28.8	21.9
Attempted failure	18.6	37.5
Pending	0.0	0.0
Conditional Fidelity	92.2	93.0
Fidelity evidence coverage	23.3	23.4
Conditional Degree estimate	48.5	37.7
Degree lower bound	11.3	8.8
Degree upper bound	51.6	39.9

3 Claim-first evaluation

3.1 Fixed denominator and final dispositions

The frozen cohort contains 100 papers, 3,933 Claim Checks and 1,427 authored Claims. Of the Claims, 1,412 have one or more numerical Checks and 15 do not. Mapping coverage is 100% by construction and validation: every Check has exactly one direct Claim and no Check is removed because it failed, was blocked or lacked a score.

Each Check receives exactly one of four evidence-derived dispositions:

1. **Successfully reproduced**: independent output and the required scientific evidence exist.
2. **Objectively blocked**: execution cannot be completed because a necessary publication input, uniquely defined observable, benchmarked compute resource or external dependency is unavailable.
3. **Attempted but not reproduced**: a scientifically meaningful implementation was run, but the Claim was not recovered and the failure does not meet the objective-blocker standard.
4. **Pending**: the evidence is not yet sufficient for any terminal disposition.

The distinction is causal rather than cosmetic. An objective blocker requires a documented direct cause, a root cause and evidence that more ordinary local iteration cannot resolve it. A code error remains an agent-side failure or repair task; it is never relabelled as an external blocker.

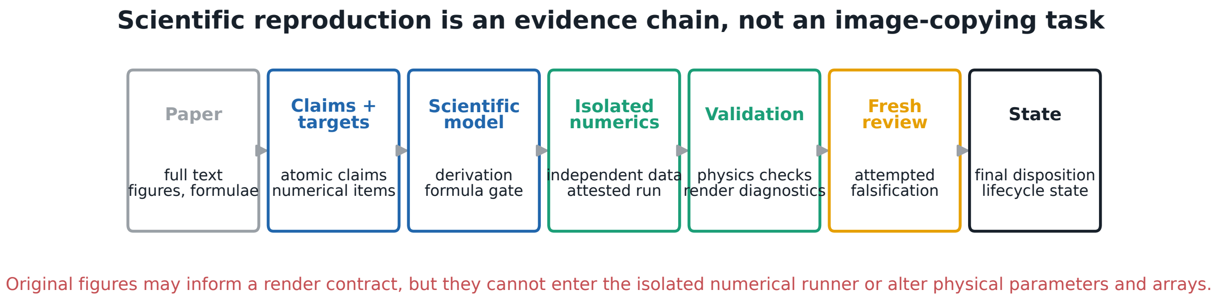

Figure 1: **Evidence flow in PRAgent.** Full-paper understanding produces scientific Claims and atomic Checks. Formula cards and parameter provenance define executable Targets. An isolated numerical runner generates data and physics checks without source-image access. A separate constrained rendering channel may inspect the publication only to align presentation. Fresh-context review attempts to falsify the Claims before authoritative state and aggregate metrics are derived.

3.2 Fidelity and Reproduction Degree

Fidelity is calculated only for a successfully reproduced Check whose every linked Target has a passed, lifecycle-eligible comparison under the `scientific_region_v1` contract. For target t with R_t declared scientific regions, the target score is

$$F_t = \frac{1}{R_t} \sum_{r=1}^{R_t} f_{tr}, \qquad 0 \le f_{tr} \le 100, \tag{1}$$

where f_{tr} is the registered pixel-similarity score for independently rendered output and the corresponding source region. The Check score is the mean of its target scores. If any required target lacks trusted comparison evidence, the Check receives no Fidelity; missing values are never guessed.

For Claim c with n_c Checks, successful coverage is

$$C_c = \frac{1}{n_c} \sum_{i=1}^{n_c} \mathbb{1}(d_{ci} = \text{reproduced}). \tag{2}$$

Corpus-level Claim coverage, blocker rate, attempted-failure rate and pending rate are macro-averages over the 1,412 numerical Claims. This prevents a paper with many supplementary panels from dominating the headline result.

For communication we define an optional product

$$D_{\text{repro}} = C_{\text{succ}} \times \frac{F_{\text{cond}}}{100}, \tag{3}$$

where C_{succ} is successful Claim coverage in per cent and F_{cond} is conditional Fidelity on the measured successful mass. Equation 3 is a conditional plug-in estimate, not an imputed full-corpus score. We therefore also report strict bounds: unmeasured successful Fidelity is set to 0 for the lower bound and 100 for the upper bound. The two raw components and Fidelity evidence coverage must always accompany the product.

3.3 Headline results

At Check level, 2,068 of 3,933 objects were successfully reproduced, 1,134 were objectively blocked, 731 were attempted without successful reproduction and 0 remained pending. Thus final resolution reached 100.0%. At Claim level, final resolution was 100.0% and successful coverage was 40.5%.

Trusted scientific-region comparisons exist for 23.4% of successful Claim mass, spanning 146 numerical Claims. On this explicitly conditional subset, Fidelity was 93.0/100. The Claim-level Degree plug-in estimate is 37.7/100, while the data-supported interval is 8.8–39.9/100. This interval is central to the interpretation: the measured comparisons show that reproduced scientific objects can be rendered with high agreement, but current evidence does not justify assigning the same Fidelity to every successful Claim.

Across papers, mean and median successful Claim coverage were 56.2% and 58.6%, respectively. Conditional paper Fidelity was available for 32 papers, with mean 93.4/100 and median 93.7/100. Conditional paper Degree was reportable for 44 papers; its mean was 50.0/100 and median 57.0/100. Figure 2 shows the full distributions rather than a single favourable aggregate.

4 What the system can and cannot reproduce

The strongest demonstrated regime is equation-defined computation for which the paper specifies parameters and the intended observable. Failures become harder to adjudicate when the publication omits a seed, finite-size selector, fit weighting, raw experimental input or an exact instance identity. The published ledgers preserve these causes at Check level, including code-fault status, so the aggregate can be decomposed without reading case narratives.

Scientific reproduction can also reveal possible paper problems. Such a label is deliberately conservative: a fresh reviewer must reproduce the discrepancy, exclude known implementation defects and identify the affected Claim and parameter scope. A paper-error candidate is therefore a review outcome, not a claim of misconduct or a replacement for author correspondence.

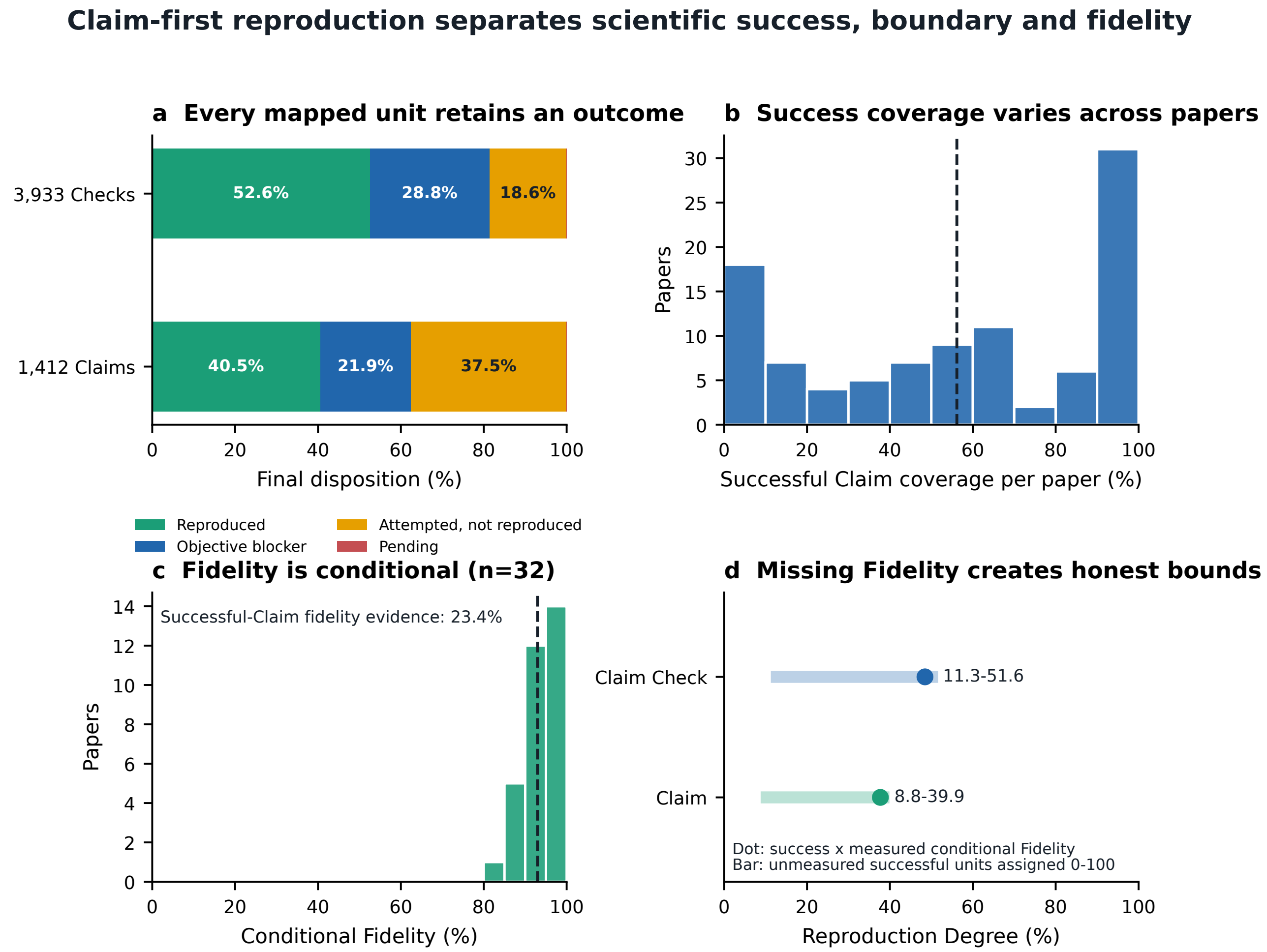

Figure 2: **Claim-first outcome and fidelity audit.** (a) Final Check- and Claim-level dispositions. (b) Distribution of successful Claim coverage across papers. (c) Conditional Fidelity for papers with trusted comparison evidence; absent evidence is not assigned a score. (d) Conditional Reproduction Degree and its non-imputation interval. Exact values and the full 100-paper ledger are provided as source data and Supplementary Tables.

5 Discussion

5.1 What has been demonstrated

The audit supports three conclusions. First, an agent can construct a broad claim-to-evidence graph across 100 heterogeneous physics papers: all 3,933 Checks are mapped, and every numerical result remains in the denominator regardless of outcome. Second, the agent can successfully reconstruct a substantial fraction of the scientific Claims from equations and publication evidence rather than source pixels. Third, it can expose its own boundary: blockers, attempted failures, pending evidence and missing Fidelity are individually enumerable rather than hidden behind a success-only gallery.

The results do not show autonomous reproduction of arbitrary research. The cohort is curated rather than sampled from all physics; difficulty varies; some published inputs are unavailable; and a passed scientific result is not identical to an independently replicated experiment. The public artifact should therefore be read as an auditable capability study and a reusable evaluation method, not as a universal success rate for physics.

5.2 Why pixels remain useful but secondary

Pixel agreement is valuable because a paper figure compresses scientific and presentational choices: curve shape, crossing position, extrema, relative scale and visible uncertainty. It is nevertheless unsafe as the first criterion. A high score can be obtained by copying source pixels, while a scientifically correct result can lose points through fonts, antialiasing or aspect ratio. The two-channel design resolves this tension. Independent numerics establish the scientific object; constrained rendering then measures how faithfully that object is communicated. Only the declared scientific regions enter Fidelity.

5.3 Why the decomposition matters

A single average obscures qualitatively different outcomes. Assigning a zero pixel score to an unavailable experimental data set would falsely describe an input boundary as a bad calculation. Conversely, averaging only the best plots would reward narrow cherry-picking. Successful coverage answers “how much was reproduced?” Conditional Fidelity answers “how close

Table 2: **Operational capability boundary.** The classification concerns what the present evidence can support, not whether a field is intrinsically reproducible.

Regime	Demonstrated capability	Present limitation
Analytic and low-dimensional numerics	Reconstruct equations; solve spectra, dynamics and phase diagrams; test normalization, symmetries, limits and scaling relations	Ambiguous conventions or unpublished selectors can leave several paper-faithful implementations without a unique adjudication
Moderate simulation and optimization	Implement independent solvers, sweeps and constrained optimization; freeze numerical outputs before rendering	Paper-scale grids, stochastic ensembles or long optimization campaigns may exceed benchmarked local or available accelerator budgets
Figure reproduction	Render frozen scientific arrays and compare declared scientific regions; diagnose layout separately	Fidelity is unavailable when the source region, generated region or lifecycle provenance is incomplete
Pure theory	Verify derivations, identities, limiting cases and invariants without forcing them into a pixel metric	A theorem without a tractable independent proof contract may remain partial rather than become a numerical success
Experiment-dependent Claims	Reanalyse accessible public data when inputs are complete and licensed	Missing raw data, calibration, device state or exact sample identity is an objective input boundary, not a numerical Fidelity value

were the successfully reproduced and validly compared objects?" Evidence coverage answers "how much of that success was actually scored?" Degree is useful only when these three answers remain visible.

6 Limitations

The study has six principal limitations. First, paper selection was purposive and the cohort should not be treated as a random sample of physics. Second, Claim granularity is authored and audited but cannot eliminate all judgement; splitting or merging Claims changes macro-averages. Third, the fidelity metric is pixel based and does not yet use a universal observable-space distance across all plot types. Fourth, Fidelity evidence remains incomplete, which is why the reported Degree is conditional and bounded. Fifth, fresh-context review reduces self-confirmation but is not equivalent to review by domain experts or paper authors. Sixth, some blockers may become solvable if authors release missing inputs or greater compute becomes available; their disposition is evidence- relative and versioned, not permanent.

7 Methods

7.1 Cohort freeze and clean-snapshot audit

The denominator was generated from a clean Git checkout at revision `09ab3fb65c5dc2d2cf05a4fbf3c8a42b4d4de5f5`. Working-tree changes and runtime caches were excluded. Publication files omitted from Git for licensing reasons were admitted only when their hashes matched the already frozen fresh-review manifests. The cohort manifest fixed exactly 100 paper identities before aggregation. The audit script loaded each case's authored Claim map and authoritative reproduction state, rejected duplicate identities, required one direct Claim per Check and verified that the disposition counts exhausted the fixed denominator.

7.2 Disposition derivation

Where an authoritative item disposition exists, it is used directly. Otherwise the Check is projected from all linked Target dispositions. A protocol-valid, current fresh review may close an older Target loop only when an analytic or attested numerical path exists and a reproduction-code defect has been excluded. Confirmed external blockers retain precedence. For the remaining Targets the ordering is conservative: pending precedes attempted failure, attempted failure precedes objective blocker, and only a nonempty set of entirely reproduced Targets makes the Check reproduced. Direct cause, root cause and code-fault status are projected for diagnosis but do not create a success or blocker by themselves.

7.3 Fidelity evidence contract

A score enters the metric only if (i) the top-level pixel-evidence record passed and uses schema version 2 or newer; (ii) the authoritative Target pixel state passed; (iii) the Target comparison contract and render acceptance both passed; (iv) policy version was `scientific_region_v1`; (v) the comparison role was `primary_scientific_region`; (vi) lifecycle eligibility was true; (vii) source-crop provenance was nonempty; and (viii) generated-crop provenance was `independent_render`. Scores must be numeric and lie in 0–100. A reproduced Check linked to an unscored Target remains reproduced but its Fidelity is missing. The pipeline contains no imputation path.

7.4 Aggregation

Check-level rates use all 3,933 Checks. Within each numerical Claim, rates are first calculated over its Checks. Corpus Claim rates are then the unweighted means across 1,412 Claims. Paper rows repeat the same aggregation over Claims, giving each Claim equal weight within a paper. Conditional Fidelity uses only successful mass with valid scores. Strict Degree bounds propagate unmeasured successful mass as 0 or 100, respectively. The complete equations, per-paper rows and machine-readable Check and Claim ledgers are included in the Supplementary Information.

7.5 Claims without numerical Checks

The 15 Claims without numerical Checks are never assigned fabricated pixels. Pure-theory Claims use explicit derivation, formula- provenance or invariant gates and are labelled verified, partial or unresolved. Experimental and meta-level Claims that lack an independent numerical object are explicitly excluded from numerical Fidelity. Their dispositions remain in a separate table so that exclusion cannot be mistaken for success.

7.6 Selective reruns

Reruns were reserved for pending Targets whose code and parameters were ready and whose missing evidence could plausibly be resolved by available local or A100 compute. A Target was not rerun merely to replace absent publication information. After fresh-review adjudication the frozen numerical rerun queue contained zero Targets, so no additional calculation was used to manufacture a clean headline. The remaining 346 Targets in the optional pixel-evidence queue require local rendering evidence, not new physics arrays.

7.7 Reproducible analysis and figures

One script produces the Check, Claim, nonnumeric-Claim and paper ledgers, the summary JSON, filtered blocker/failure tables and manuscript macros. A second script generates the Supplementary tables. Figure 2 is generated from the committed paper ledger and exports PDF, SVG, PNG and TIFF with panel-level CSV source data. Unit tests verify denominator exhaustion, non-imputation, score eligibility, aggregation bounds and deterministic table generation.

8 Conclusion

PRAgent turns paper reproduction into a measurable scientific workflow. In a fixed cohort of 100 physics papers, 3,933 atomic Checks map to 1,427 Claims with complete structural traceability. At Claim level, 40.5% was successfully reproduced and 100.0% reached a final disposition. For the trusted measured successful Claim mass, conditional Fidelity was 93.0/100; the optional Degree estimate was 37.7/100 with bounds 8.8–39.9/100. The central contribution is the decomposition itself: what was reproduced, what was externally impossible, what was tried and failed, and what visual evidence is genuinely known are all reported separately. This establishes a strong but falsifiable baseline for claim-level AI scientific reproduction.

Data and code availability

The public RunThePaper release contains the 100-paper cohort and the generated Check, Claim and paper ledgers, summary tables, figure source data and case artifacts permitted for redistribution. The corresponding harness, validation scripts and manuscript sources are versioned in PRAgent. The audit input is the Git revision recorded above; later case improvements must produce a new versioned snapshot rather than silently changing this result. Paper materials remain subject to their source licences.

References

[1] Steven N. Goodman, Daniele Fanelli, and John P. A. Ioannidis. What does research reproducibility mean? *Science Translational Medicine*, 8(341), 2016. doi: 10.1126/scitranslmed.aaf5027.

[2] Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624(7992): 570–578, 2023. doi: 10.1038/s41586-023-06792-0.

[3] Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D. White, and Philippe Schwaller. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6(5): 525–535, 2024. doi: 10.1038/s42256-024-00832-8.

[4] Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The AI scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*, 2024. URL https://arxiv.org/abs/2408.06292.

[5] Zachary S. Siegel, Sayash Kapoor, Nitya Nadgir, Benedikt Stroebl, and Arvind Narayanan. CORE-Bench: Fostering the credibility of published research through a computational reproducibility agent benchmark. *arXiv preprint arXiv:2409.11363*, 2024. URL https://arxiv.org/abs/2409.11363.

[6] Yanzheng Xiang, Hanqi Yan, Shuyin Ouyang, Lin Gui, and Yulan He. SciReplicate-Bench: Benchmarking LLMs in agent-driven algorithmic reproduction from research papers. *arXiv preprint arXiv:2504.00255*, 2025. URL https://arxiv.org/abs/2504.00255.

[7] Giulio Starace, Oliver Jaffe, Dane Sherburn, James Aung, Jun Shern Chan, Leon Maksin, Rachel Dias, Evan Mays, Benjamin Kinsella, Wyatt Thompson, Johannes Heidecke, Amelia Glaese, and Tejal Patwardhan. PaperBench: Evaluating AI's ability to replicate AI research. *arXiv preprint arXiv:2504.01848*, 2025. URL https://arxiv.org/abs/2504.01848.

[8] Gyeongwon James Kim, Alex Wilf, Louis-Philippe Morency, and Daniel Fried. From reproduction to replication: Evaluating research agents

with progressive code masking. *arXiv preprint arXiv:2506.19724*, 2025. URL https://arxiv.org/abs/2506.19724.

[9] Christine Ye, Sihan Yuan, Suchetha Cooray, Steven Dillmann, Ian L. V. Roque, Dalya Baron, Philipp Frank, Sergio Martin-Alvarez, Nolan Koblischke, Frank J. Qu, Diyi Yang, Risa Wechsler, and Ioana Ciuca. ReplicationBench: Can AI agents replicate astrophysics research papers? *arXiv preprint arXiv:2510.24591*, 2025. URL https://arxiv.org/abs/2510.24591.

[10] Bang Nguyen, Dominik Soós, Qian Ma, Rochana R. Obadage, Zack Ranjan, Sai Koneru, Timothy M. Errington, Shakhlo Nematova, Sarah Rajtmajer, Jian Wu, and Meng Jiang. ReplicatorBench: Benchmarking LLM agents for replicability in social and behavioral sciences. *arXiv preprint arXiv:2602.11354*, 2026. URL https://arxiv.org/abs/2602.11354.

[11] Shi Qiu, Junyi Deng, Yiwei Deng, Haoran Dong, Jieyu Fu, Mao Li, Zeyu Li, Zhaolong Zhang, Huiwen Zheng, et al. PRBench: End-to-end paper reproduction in physics research. *arXiv preprint arXiv:2603.27646*, 2026. URL https://arxiv.org/abs/2603.27646.

[12] Tingjia Miao, Wenkai Jin, Muhua Zhang, Jinxin Tan, Yuelin Hu, Tu Guo, Jiejun Zhang, Yuhan Wang, Wenbo Li, Yinuo Gao, et al. PRL-Bench: A comprehensive benchmark for evaluating the capabilities of LLMs in frontier physics research. *arXiv preprint arXiv:2604.15411*, 2026. URL https://arxiv.org/abs/2604.15411.